

EDUBEHAVIORS: ASSERTION-BASED SCHEMAS FOR AUDITABLE CODING OF EDUCATIONAL DIALOGUES

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ABSTRACT. Large language models have allowed the rapid deployment of pedagogical annotations corresponding to constructs of interest, allowing a natural language interface for generating classifications on a conversational dataset. However due to the opaque nature of LLM reasoning, we have no verifiable, mechanistic insight into *why* a model chose a label for an utterance. We introduce the *EduBehaviors* framework, an interpretable, scalable approach to annotating educational data that uses LLMs to measure repeated observable behaviors relevant to many constructs of interest and then learns a classifier for the construct based on these observable behaviors. We evaluate the framework on the TalkMoves dataset, predicting the Teacher TalkMoves labels. Our best configuration results in a macro-F1 of 0.673 and 0.688 Cohen’s κ , proving competitive with direct prompting approaches. In addition, we release *EduBehaviors Toolkit*, two tools allowing researchers to operationalize the *EduBehaviors* framework in their own data.

1. INTRODUCTION

Large language models (LLMs) are increasingly being used for text and discourse annotation, with studies showing that they can match or exceed crowdworker performance on some classification tasks at substantially lower cost [7, 33]. This development creates new opportunities for large-scale, theoretically grounded analysis of conversational data, especially in education research, where teacher-student talk has long been treated as an important source of evidence about instructional quality and student learning [5, 17].

Recent work has applied LLMs to classroom and tutoring discourse constructs at scale [15, 18], extending earlier supervised approaches [6, 24] and informing analyses of emerging human-AI tutoring systems [29].

Using LLMs for annotation still requires researchers to define constructs, develop coding procedures, and validate that model outputs capture the intended distinctions [8, 19, 22]. These activities require expertise and iterative examination of examples, adding to the work involved in producing usable measures. When an LLM assigns a construct label directly, disagreements with human coders can be difficult to diagnose: they may reflect model error, ambiguous evidence, or insufficiently specified construct definitions. Prompt refinement is especially challenging when reference labels are sparse or unavailable, and existing optimization methods typically require explicit feedback signals or labeled development data [9, 20].

Evidence from educational discourse illustrates these annotation challenges. Using the Talk-Moves dataset [24], Vanacore and Kizilcec [27] found that the strongest tested model-prompt combination achieved only moderate overall agreement with expert coders ($\kappa = 0.58$), despite

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using a human-validated coding manual and few-shot examples. Related studies also report difficulties with abstract constructs and judgments requiring nuanced interpretation [14, 25]. These findings motivate evaluating both the accuracy of construct labels and the intermediate judgments used to produce them.

We introduce EduBehaviors, a framework that makes these coding decisions available for inspection by separating the identification of observable behaviors from their combination into construct labels. Researchers develop explicit statements about conversational behavior, which we call assertions. LLMs annotate each assertion as true or false, and a specified rule or trained statistical model combines these values into a construct label. This structure allows researchers to examine the evidence used in a classification, revise individual assertions, and assess how those assertions contribute to the final label.

1.1. Contributions. Our primary contribution is the *EduBehaviors* framework for making the coding of educational conversations inspectable and revisable. The framework exposes both the behavioral judgments used as evidence and the rule or model that maps those judgments to a construct label. It also supports reuse: an assertion annotated once in a corpus can inform multiple construct labels, and assertion annotations can be used to train smaller classifiers for subsequent annotation. These features are intended to reduce repeated development and annotation work. Their use still requires evidence that the assertions and resulting labels capture the intended constructs in the target context.

We evaluate the framework on the expert-labeled TalkMoves dataset [24], comparing its predictive performance with direct LLM prompting and examining agreement across LLM annotators. We also release two open-source tools: EduBehaviors-Studio, an interface for developing and refining assertion-based schemas, and EduBehaviors-kit, an encoder-based annotation pipeline.

2. RELATED WORK

2.1. Developing annotation criteria through theory and data. Operationalizing educational constructs requires connecting theoretical definitions with evidence that can be identified in conversational data. Research on LLM-assisted codebook development approaches this task through a combination of educational theory, empirical examples, and human refinement [32]. Complementary work on concept induction helps researchers discover patterns in unstructured text and articulate explicit criteria for identifying them [13]. Recent systems support mixed-initiative codebook discovery [30] and orchestrated AI-assisted annotation [10]. Together, these approaches position LLMs as resources for developing and revising operationalizations through engagement with both theory and data.

This development process is inherently iterative: applying criteria to examples can expose ambiguities and reshape the criteria themselves. Research on LLM-assisted evaluation describes this interaction as *criteria drift*, emphasizing that defining criteria and examining their applications are interdependent activities [23]. These ideas inform both the curation of corpus-derived assertions and the researcher-guided refinement of schema-derived assertions in EduBehaviors.

2.2. Decomposition and interpretable construct modeling. Once annotation criteria have been articulated, a related question is how to organize them into an inspectable prediction process. Decomposed prompting and fine-grained evaluation share the principle that complex tasks or broad judgments can be separated into components that are easier to specify, assess, and refine individually [11, 31]. This principle motivates treating behavioral assertions as distinct annotation targets.

Concept bottleneck models connect these intermediate judgments to final predictions by making human-understandable concepts an explicit part of the predictive process [12]. Extensions to text classification further demonstrate how LLMs can help identify and measure such concepts [16]. Together, decomposition and concept-based prediction provide foundations for separating the identification of behavioral evidence from its interpretation as an educational construct. This separation also connects prediction to the iterative development process described above, making the intermediate criteria available for researcher inspection and revision.

2.3. Reusable models and educational research infrastructure. Making annotation components reusable requires ways to translate researcher expertise into supervision at scale. Snorkel establishes a precedent through programmatic labeling functions whose noisy outputs are combined into training labels [21]. Related work on distillation uses LLM-generated explanations or labeled examples to train smaller classifiers [1, 3]. Together, these approaches connect the specification of supervision with efficient model deployment, informing EduBehaviors’ use of LLM-generated assertion labels to train behavioral encoders. Whereas Snorkel combines alternative labels for the same target, EduBehaviors assertions describe distinct behaviors whose predictions serve as features for construct modeling.

Shared infrastructure makes such models available beyond the studies in which they were developed. Edu-ConvoKit supports this goal by integrating educational conversation analysis tools, including established models for focusing questions, accountable talk moves, and conversational uptake [28]. It provides complementary resources to EduBehaviors’ emphasis on reusable behavioral assertions. Viewed together, these literatures connect researcher-guided operationalization, interpretable prediction, and efficient deployment—the three concerns linking EduBehaviors-Studio and EduBehaviors-kit.

3. THE EDUBEHAVIORS FRAMEWORK

Consider a set of utterances $\mathcal{U}$ and the task of annotating each utterance $u \in \mathcal{U}$ with a label $L \in \mathcal{L} = \{1, \dots, k\}$. Now, let an **assertion** A be a binary-valued function $A : \mathcal{U} \rightarrow \{0, 1\}$ on the set of utterances. Then, a **decomposable schema** $S = (\mathcal{A}, R)$ is a set of assertions $\mathbf{A} = (A_1, \dots, A_n)$ along with a rule $R : \{0, 1\}^n \rightarrow \mathcal{L}$ mapping every combination of assertion values into the label space. Taken together, an assertion set and rule can be composed to form a map $M : \mathcal{U} \rightarrow \mathcal{L}$ into the label space defined as $M(u) = R(A_1(u), \dots, A_n(u))$. The map M corresponding to a schema S can then be used to predict a label $\ell \in \mathcal{L}$ for every utterance $u \in \mathcal{U}$. As formulated here, a decomposable schema is an example of a *concept bottleneck model*, with concept vector $\mathbf{A}$ and task function R [24]. In this section, we describe the *EduBehaviors* framework: a modular, transparent, cost-sensitive approach to annotating a construct in educational discourse data.

3.1. Generating assertions. For our decomposable schema S to be interpretable, we need its assertions to correspond to human-readable, directly observable, aspects of utterances. For example, “utterance is spoken by a student,” or “utterance includes a number” are viable aspects that can be directly annotated in the corpus using a language model. The behaviors encoded by assertions can be human-prescribed or LLM-generated. The EduBehaviors framework prescribes two steps to generating assertions: generate assertions that describe the corpus of annotation, then generate assertions that decompose the construct of interest.

Corpus-derived assertions are generated directly from the corpus, with no grounding in a specific downstream task. In a pedagogical dialogue setting, this may include assertions describing an utterance’s pedagogical function (“utterance poses a new exercise”), intended

recipient ("utterance is directed at a student"), relationship to other utterances ("utterance corrects a mistake"), and any other dichotomous aspect of an utterance. The corpus-derived assertions form the construct-independent set of assertions that can be used to annotate *any future construct of interest* on the corpus.

Construct-derived assertions are generated directly from a description of the construct of interest with no grounding in the corpus. These assertions should correspond to observable aspects corresponding to one or more construct labels. For example, if the task of interest is identifying utterances expressing curiosity in tutoring sessions, a natural assertion might be "utterance asks a question" or "utterance expressed interest in lesson." These assertions may be used as evidence that an utterance expresses curiosity, but break the concept down beyond the more subjective annotation "utterance expresses curiosity."

For assertions to be useful, their annotation in the dataset should be minimally ambiguous: any two unprepared human annotators should highly agree on which utterances an assertion should be true-valued. While assertion annotations can be reviewed directly, managing a library of reusable assertions requires effective triage of assertions for rewriting. To get a proxy for assertion faithfulness, we annotate each assertion using a library of multiple LLM annotators, then calculate inter-rater reliability for each assertion. Although models may share systematic errors even in this scenario [4], our approach makes measurement judgment more inspectable and allows us to identify cases in which assertions may need to be written differently or discarded if they fail to pass a minimum agreement bar (e.g. Cohen's $\kappa > 0.5$).

3.2. Defining a rule. Once the observable aspects of interest have been extracted from the corpus and construct definition and collected and operationalized into a set of assertions $\mathcal{A} = \{A_1, \dots, A_n\}$, the rule $R : \{0, 1\}^n \rightarrow \mathcal{L}$ must be decided. However, since our rule only operates on the binary assertion values, this amounts to a classification task with binary covariates. As such, transparent linear classifiers such as logistic regression may be used to maximize interpretability of the resulting schema. Otherwise, more flexible procedures like a random forest or fine-tuned neural network might be used to capture complex relationships between assertions and improve predictive accuracy.

Because the rule definition step is a standard tabular supervised learning problem, our framework has some desirable properties not present in direct LLM annotation:

- **Calibration:** Calibrated classification probabilities are not straightforward to derive from LLM annotations [2], whereas our framework's classifier may be calibrated with standard techniques. This enables utterance-level triage of uncertain predictions.
- **Tunability:** Precision and recall are competing goals for a classifier that may each be desired in different situations. Prompt-based LLM annotation gives no reliable control over this property of the resulting classifications, while many supervised classification functions have thresholds directly controlling the precision-recall tradeoff.

The *EduBehaviors* framework described above outlines a procedure for annotating educational conversational data that incorporates the natural language interface of direct LLM prompting into a supervised learning framework. By doing so, we increase the interpretability and reproducibility of our classifications and produce LLM artifacts that can be re-used across many tasks.

4. EDUBEHAVIORS TOOLKIT

In this section we describe the EduBehaviors Toolkit, consisting of **EduBehaviors-Studio**, an LLM-assisted UI for generating construct-derived assertions, and **EduBehaviors-kit**, a Python package allowing researchers to apply the EduBehaviors framework to their own dataset to generate annotations.

4.1. EduBehaviors-Studio. EduBehaviors-Studio¹ is a self-hosted, bring-your-own-key web-app allowing researchers to generate a set of construct-derived assertions, annotate them on a dataset of choice, then iterate on these assertions until they adequately reflect a construct of interest. The app is defined by a single workflow:

- (1) **Describe** your construct of interest, its possible values, and whether it manifests in tutor utterances, student utterances, or both.
- (2) **Approve**, reject, or edit a set of LLM-generated, construct-derived assertions based on your description.
- (3) **Review** utterances to see how whether the construct-derived assertions’ implied mapping aligns with your intuition of the construct. Also, review the values annotated in the dataset to check for assertion faithfulness.
- (4) **Revise** assertions or their mapping into a final construct label based on disagreements observed in the review round.
- (5) **Repeat** until the set of construct-derived assertions can consistently identify cases of your construct of interest in a conversational dataset.

Our review process centers our distinction of **definitional errors** from **predictive errors**. When encountering a construct classification that a reviewer disagrees with, the reviewer can observe the assertions that fired for a given utterance. If the reviewer is in agreement with the LLM’s judgement across those utterances yet still disagrees with the final label, this exposes a gap in the measure’s definition rather than a flaw in the LLM’s annotation. At this point, the reviewer can rephrase an existing assertion, add a new assertion, or revise the aggregation rule. If the rule R is the only change, the counterfactual annotation can be immediately derived with no additional LLM calls. If there is a change to assertions, then the additional costs corresponding to schema refinement are reduced since existing assertions need not be re-annotated. These properties give actionable next steps when labels diverge from reviewer judgement beyond just “make the prompt better.”

4.2. EduBehaviors-kit. We introduce **EduBehaviors-kit**, a combination of (I) an open-source, pip-installable Python package accessible on GitHub² and PyPI³, and (II) a collection of pretrained open-weights encoder models and training data accessible on HuggingFace⁴, in order to provide an easy-to-use system for researchers to apply the methods described in this paper to their own work. **EduBehaviors-kit** was created with two main goals in mind:

Democratization of educational text classification methods. **EduBehaviors-kit** is intended to make lightweight behavior annotation and model training as simple as possible for researchers with little to no experience with language model-based text annotation or classification. The package allows novice users to easily replicate a pipeline similar to the one described in this paper; annotating a dataset with all available assertion models and counts of default words,

¹<https://github.com/scale-nssa/edubehaviors-studio/>

²<https://github.com/scale-nssa/edubehaviors-kit/>

³<https://pypi.org/project/edubehaviors-kit/>

⁴<https://huggingface.co/collections/StanfordSCALE/assertions>

splitting into train/test sets, training and evaluating an interpretable classification model is as simple as calling `ClassificationPipeline(data, words="all", assertions="all")`.

Consistent, efficient, and open models. By using pretrained encoders to annotate data instead of proprietary language models, we can improve both the reliability and cost of annotation. As such, we provide pretrained classifiers for 49 of the bottom-up assertions used in this paper on our HuggingFace repository. SetFit [26] models are trained on LLM annotations of tutor utterances from a subset of the TalkMoves dataset; training data, performance metrics, and model hyperparameters are publically available. These lightweight models can be run on a laptop, or cheaply on computing systems. Unlike proprietary language models, encoders also allow for fully reproducible annotation results. By sharing model weights, data and training information, we encourage other users of this work to create their own assertion classifiers or improve upon ours.

4.2.1. *Package structure.* The package contains four main objects:

- (1) `WordAnnotator`: Annotates data with counts or appearances of words in a list.
- (2) `AssertionAnnotator`: Annotates data using the available assertion encoder models
- (3) `standard_classifier`: Instantiates a logistic regression classifier with reasonable hyperparameters similar to that used in the analysis for this paper.
- (4) `ClassificationPipeline`: Creates and runs an end-to-end pipeline; annotates data, creates train/test sets, and trains a classification model.

Full documentation of these objects, including lists of available assertion classifiers and default words, and examples of how to create an end-to-end classification pipeline, are available on GitHub.

5. PERFORMANCE

In this section, we will demonstrate that the *EduBehaviors* framework for utterance annotation can be directly applied to tasks of pedagogical interest. While previous sections have established that our framework’s predictions have unique desirable properties, it remains to be seen whether its classifications can match the accuracy of direct LLM prompting.

5.1. **Task.** The TalkMoves Dataset is a set of 566 human-generated transcripts of mathematics lessons taking place in real classrooms [24]. These transcripts were then annotated with the TalkMoves coding manual which describes 10 “talk moves,” discursive techniques of pedagogical interest used by either the teacher or student. These

In our experiment we focus solely on the seven labels described in the Teacher TalkMoves coding manual. To get an unbiased estimate of our classifier performance without annotating the whole dataset we sample 10 sessions, consisting of 3,217 total teacher utterances to annotate. The metrics of interest are Macro-F1 on the gold labels over the 7 classes as well as inter-rater reliability with the gold annotations as measured by Cohen’s κ . We will estimate these metrics using leave-one-session-out cross validation by averaging performance on the unseen test set.

The TalkMoves dataset is commonly used as a benchmark for prediction in education settings, allowing us to understand our results in light of existing benchmarks on the dataset. Fine-tuned RoBERTa-base performance as presented in the original paper maxed out at 0.76 macro-F1 with no agreement metrics reported. LLM-prompted performance lies distinctly below the encoder performance; Vanacore and Kizilcec [27] evaluated the performance of LLMs from three Frontier labs and found a maximum macro-F1 of 0.61 and maximum Cohen’s κ of 0.58, associated with directly prompting Claude Opus 4.5 with the TalkMoves coding manual and few-shot examples.

5.2. Experimental Setup. To benchmark the performance of the EduBehaviors framework on the Teacher TalkMoves task, we generate both corpus-derived and construct-derived assertions once, annotate them with a variety of language models, then learn an L1-regularized, logistic regression map from the assertion-values into the label space. We describe the details of this process below.

Assertion generation is once-off and shared across all models. For corpus-derived assertion generation, we prompted Claude Fable 5 with TalkMoves transcripts and the task of describing in general terms *what is happening* in each utterance. Across many transcripts, this resulted in thousands of assertions. We took two approaches to de-duplication. First, we fed the utterance-level annotations into the same model to generate 61 assertions summarizing the range of behavior described in the utterance-level annotations. In addition, we conducted simple lemmatization (plural and article removal) across the utterance-level annotations, then took the top 50 resulting assertions. After a human-led de-duplication of these two lists, our final list contains 74 corpus-derived assertions. For construct-derived assertions, we prompted Gemini 3.1 Pro with the Teacher TalkMoves section of the TalkMoves coding manual and directly generated 48 assertions with no inclusion of transcripts. Finally, as an automatic source of corpus-derived assertions, we ranked the top 100 words present in a disjoint set of transcripts, then included 100 additional corpus-derived assertions of the form “The utterance contains the word X.” In total, this resulted in 221 total assertions to annotate and be used for classification.

Assertion annotation was conducted with a panel of five language models: **Claude Opus 4.8**, **Gemini 3.6 Flash**, **Gemini 3.5 Flash-Lite**, **GPT 5.6 Luna**, and **GPT OSS 120B**. These represent a range of providers, sizes, and includes an open-weights model (GPT OSS). After annotation, cross-model agreement was calculated using Krippendorff’s α for each assertion.

Classification was done by fitting an L1-penalized logistic regression to the final labels based on the gold Teacher TalkMoves annotations. In particular, we fit 7 one-way classifiers corresponding to each possible Teacher TalkMoves label, including “None.” Based on an independent inner-validation loop, each classifier selects a regularization penalty λ and selection threshold τ minimizing validation loss. Then, final labels are selected by taking the arg max over the resulting log probabilities divided by its selection threshold τ , to account for class imbalance. We construct ten annotators, leaving out each test session to check for accuracy, then average across performance metrics in these 10 runs.

As a consistent baseline, we predict the final labels using only the “Utterance contains word X” assertions, which are calculated automatically and annotator-independent. Then, we run each annotator in four conditions corresponding to the set of assertions included in the classification step. As a first axis, we compare the classifiers learned using only an annotator’s corpus-derived assertion values to the classifiers learned using both corpus-derived and construct-derived assertions. As a second axis, we introduce a cross-model agreement filter on assertions, only including assertions with Krippendorff’s α greater than or equal to 0.5. As such, we isolate the contributions of the separate assertion generation sources as well as the potential for our assertion faithfulness proxy to narrow the set of assertions to those with greater expected human agreement.

5.3. Results. In this section we describe the results of our assertion annotation, then the resulting classifications.

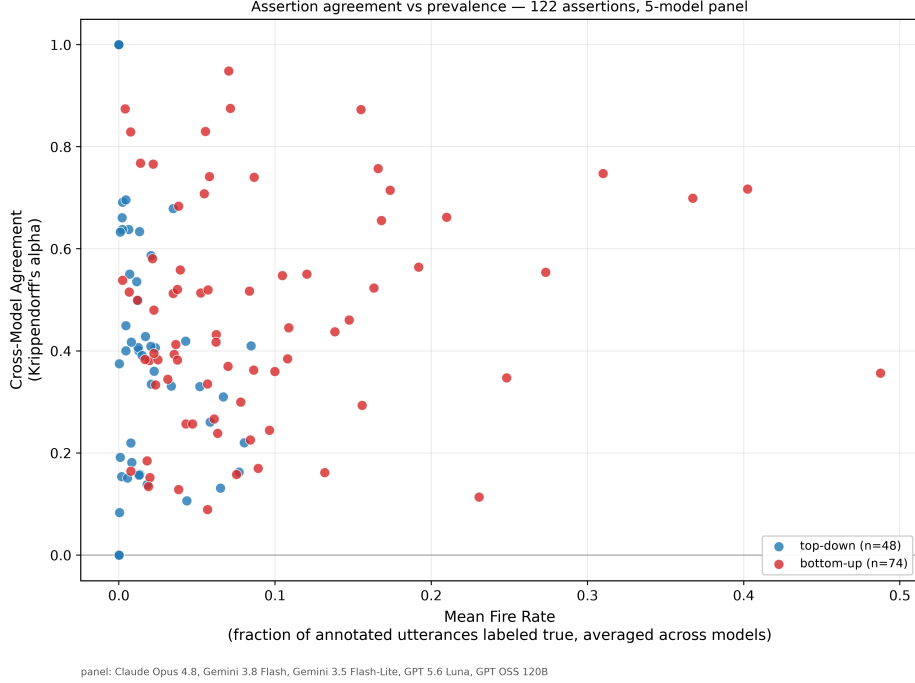


FIGURE 1. Fire rate (fraction of teacher utterances where that assertion is true) versus cross-model agreement for each assertion. Fire rate is averaged across the five annotators. Construct-derived (top-down) assertions are colored in blue while Corpus-derived (bottom-up) assertions are colored in red.

5.3.1. Annotation Results. The mean prevalence and cross-model agreement of each assertion can be seen in Figure 1. Overall, cross-model agreement is moderate (median = 0.401 and higher among the construct-derived assertions than the data-derived assertions. Fire rate also varies between the two categories, with the corpus-derived assertions appearing more than those that were construct-derived. For subsequent analysis, filtering for $\alpha \geq 0.5$ on this corpus-derived set leaves 33 of 74 corpus-derived assertions and 11 of 48 construct-derived assertions.

5.3.2. Classification Results. The best performance across the four configurations for each annotator is reported in Table 1. Overall, we find that all three assertion sources contribute to predictive accuracy as measured by Macro-F1 and Cohen's κ . Furthermore, our top three annotators perform better with the cross-model agreement filter. GPT 5.6 Luna attains the highest Macro-F1 of 0.673, and Gemini 3.6 Flash attains the highest Cohen's κ of 0.702. Individual F_1 scores are shown per label, demonstrating significant variability.

6. DISCUSSION

We developed EduBehaviors while trying to measure many educational constructs, each requiring time-consuming and costly coding and validation work. Some of that work overlapped because different constructs drew on the same conversational behaviors. These behaviors can inform measures of teaching practices and what students reveal about their thinking

Annotator	Best covariate set	Overall		Per-class F_1							
		Macro- F_1	κ	None	KET	PFA	Revo	Rest	GSTR	PFR	
GPT-5.6 Luna	words + corpus-derived + construct-derived, $\alpha > 0.5$	0.673	0.688	0.932	0.582	0.794	0.432	0.721	0.571	0.681	
Gemini 3.6 Flash	words + corpus-derived + construct-derived, $\alpha > 0.5$	0.664	0.702	0.941	0.629	0.771	0.397	0.793	0.435	0.682	
Gemini 3.5 Flash-Lite	words + corpus-derived + construct-derived, $\alpha > 0.5$	0.662	0.661	0.928	0.547	0.758	0.444	0.667	0.552	0.739	
Opus 4.8	words + corpus-derived + construct-derived	0.662	0.662	0.926	0.539	0.775	0.439	0.678	0.518	0.756	
GPT-OSS 120B	words + corpus-derived + construct-derived	0.527	0.589	0.907	0.439	0.771	0.373	0.633	0.158	0.410	
Words only (no annotator)	words only	0.339	0.328	0.822	0.225	0.628	0.000	0.066	0.290	0.345	

TABLE 1. Best covariate set per annotator, with overall and per-class classification performance. F_1 and κ scores averaged across leave-one-session out test sets on sample of 10 sessions, 3,217 gold-labeled utterances.

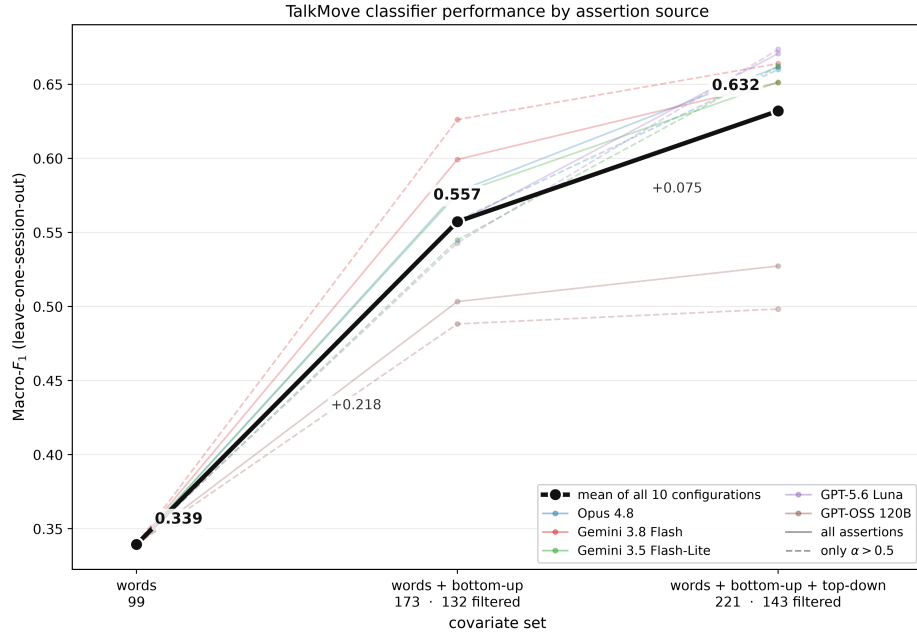


FIGURE 2. Average Macro-F1 of the TalkMoves classifiers with a growing set of assertions. Black line is an average across all runs and annotators.

and emotions. EduBehaviors organizes coding into two steps: identifying specific behaviors and combining those observations into construct labels. For example, identifying whether an educator asks a question can contribute to measures of focusing questions, funneling questions, responses to student mistakes, or relationship building. Other evidence helps distinguish these uses, including what the question asks, whether it follows a student error, and how it relates to the student’s preceding contribution. Researchers can reuse the question annotation across these measures while evaluating whether each combination of evidence captures the intended construct.

Representing this evidence as explicit assertions makes coding decisions available for review, including judgments that a codebook may leave unstated. Researchers can examine an assertion’s definition, check its application against examples from the corpus, and assess how it contributes to the final label. The approach may require annotating more individual features,

with each assertion intended to capture a behavior that annotators can recognize consistently. When disagreements arise, researchers can investigate whether the assertion needs clarification, whether it was annotated incorrectly, or whether the rule combining assertions needs revision. Whether this process reduces the number of annotation and revision rounds requires further evaluation.

In the TalkMoves evaluation, the approach generally improved performance over direct LLM prompting, although gains varied across models and metrics. It did not match the performance of the specialized classifier trained on expert-labeled TalkMoves data. These findings establish the approach’s predictive performance in one setting. The assertion annotations also provide a way to investigate how individual predictions were produced, including incorrect predictions. Agreement across models can help researchers prioritize assertions for review, but human checks remain necessary because models can share errors.

Annotated assertions can also provide training data for smaller classifiers, potentially reducing the time and computational cost of annotating larger samples. Applying these classifiers to another corpus requires checking that they identify the intended behaviors in the new setting. EduBehaviors-Studio and EduBehaviors-kit support the development and application of these annotation components. Sharing assertion definitions, annotated examples, trained classifiers, and combination rules would give researchers resources they can examine and adapt, reducing the need to develop every measure from the beginning. We hope these shared resources will make the analysis of educational conversations more accessible and help researchers build on one another’s work.

7. CONCLUSION

We introduced EduBehaviors, a framework for annotating educational conversations by identifying specific behaviors and combining those observations into construct labels. This structure allows researchers to inspect and revise individual coding decisions and reuse relevant behavioral annotations across constructs. In the TalkMoves evaluation, the approach generally improved performance over direct LLM prompting, although results varied across models and metrics and remained below those of the specialized classifier trained on expert-labeled data. EduBehaviors-Studio and EduBehaviors-kit provide tools for developing and applying assertion-based schemas. Together, the framework and tools help researchers examine how their measures are constructed and build on existing annotation work, while retaining the need to validate each measure for its intended use.

LIMITATIONS

Validation of assertion annotations. We acknowledge that validation of LLM-annotated assertions against a human-generated, golden dataset is preferable whenever possible, but recognize the challenges of creating validation sets for dozens of assertions. In the absence of reference assertion labels, cross-model agreement provides evidence of consistency across annotation systems, but cannot exclude shared errors or independently establish construct validity. We caution users of our framework relying on cross-LLM agreement as a proxy for assertion validity to audit assertion annotations even when cross-LLM agreement is high.

EduBehaviors-kit classifiers out of context. At the time of writing, all encoder-based classifiers available through the EduBehaviors-kit are based on TalkMoves corpus-derived assertions, using LLM-annotated labels on the TalkMoves corpus as training and validation.

Thus, these models may not be valid for applications that differ in significant dimensions, and their use should be carefully evaluated in such cases.

Framework flexibility and user discretion. Our framework is highly flexible and depends on the user to make sensible choices when implementing it on a new context. Our toolkit application is modeled after the framework demonstration presented in this paper as an accessible implementation, but is not intended to be prescriptive.

Performance against fine-tuned encoders. While our framework outperforms directly prompting LLMs in most of the tested cases, it strictly underperforms the original fine-tuned encoder-based classifier trained on the golden dataset annotated by experts. Thus, the investment necessary for creating such datasets may still be a worthwhile one in conditions that require a high degree of confidence in model performance.

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